

Assignment_2

February 28, 2025

```
[ ]: Name : Omkar Hulawale  
Class , Batch : TE-A , A3 Batch  
Roll No: 13165
```

```
[7]: import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt
```

```
[8]: df = pd.read_csv(r"C:\Users\hulaw\OneDrive\Desktop\DSBDA\Emp_data.csv")
```

```
[9]: df
```

```
[9]:
```

	Sr.No	Emp_ID	Emp_Salary	Emp_Marks	Emp_Join_Date
0	1	90	63773	7	01-02-2015
1	2	15	86346	6	02-02-2015
2	3	22	71456	5	03-02-2015
3	4	43	60311	7	04-02-2015
4	5	25	70284	6	05-02-2015
5	6	55	60836	9	06-02-2015
6	7	14	52276	8	07-02-2015
7	8	29	79744	6	08-02-2015
8	9	46	65538	10	09-02-2015
9	10	9	77429	8	10-02-2015
10	11	47	97998	5	11-02-2015
11	12	55	87633	8	12-02-2015
12	13	79	85773	9	13-02-2015
13	14	61	53649	10	14-02-2015
14	15	41	74034	5	15-02-2015
15	16	7	64077	10	16-02-2015
16	17	45	57106	8	17-02-2015
17	18	58	58274	7	18-02-2015
18	19	72	97271	7	19-02-2015
19	20	65	52475	5	20-02-2015
20	21	60	92853	5	21-02-2015
21	22	43	93106	7	22-02-2015
22	23	72	99773	10	23-02-2015
23	24	68	60013	10	24-02-2015
24	25	8	88503	10	25-02-2015

25	26	40	84479	8	26-02-2015
26	27	44	51156	7	27-02-2015
27	28	9	58126	10	28-02-2015
28	29	54	79824	10	01-03-2015
29	30	20	97448	5	02-03-2015

```
[10]: series = pd.notnull(df["Emp_Marks"])
```

```
[15]: print(series)
```

```
0    True
1    True
2    True
3    True
4    True
5    True
6    True
7    True
8    True
9    True
10   True
11   True
12   True
13   True
14   True
15   True
16   True
17   True
18   True
19   True
20   True
21   True
22   True
23   True
24   True
25   True
26   True
27   True
28   True
29   True
Name: Emp_Marks, dtype: bool
```

```
[17]: df.isnull()
```

```
[17]:
```

	Sr.No	Emp_ID	Emp_Salary	Emp_Marks	Emp_Join_Date
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False

3	False	False	False	False	False
4	False	False	False	False	False
5	False	False	False	False	False
6	False	False	False	False	False
7	False	False	False	False	False
8	False	False	False	False	False
9	False	False	False	False	False
10	False	False	False	False	False
11	False	False	False	False	False
12	False	False	False	False	False
13	False	False	False	False	False
14	False	False	False	False	False
15	False	False	False	False	False
16	False	False	False	False	False
17	False	False	False	False	False
18	False	False	False	False	False
19	False	False	False	False	False
20	False	False	False	False	False
21	False	False	False	False	False
22	False	False	False	False	False
23	False	False	False	False	False
24	False	False	False	False	False
25	False	False	False	False	False
26	False	False	False	False	False
27	False	False	False	False	False
28	False	False	False	False	False
29	False	False	False	False	False

```
[19]: df.notnull()
```

```
[19]:
```

	Sr.No	Emp_ID	Emp_Salary	Emp_Marks	Emp_Join_Date
0	True	True	True	True	True
1	True	True	True	True	True
2	True	True	True	True	True
3	True	True	True	True	True
4	True	True	True	True	True
5	True	True	True	True	True
6	True	True	True	True	True
7	True	True	True	True	True
8	True	True	True	True	True
9	True	True	True	True	True
10	True	True	True	True	True
11	True	True	True	True	True
12	True	True	True	True	True
13	True	True	True	True	True
14	True	True	True	True	True
15	True	True	True	True	True

16	True	True	True	True	True
17	True	True	True	True	True
18	True	True	True	True	True
19	True	True	True	True	True
20	True	True	True	True	True
21	True	True	True	True	True
22	True	True	True	True	True
23	True	True	True	True	True
24	True	True	True	True	True
25	True	True	True	True	True
26	True	True	True	True	True
27	True	True	True	True	True
28	True	True	True	True	True
29	True	True	True	True	True

```
[21]: df.dropna()
```

```
[21]:
```

	Sr.No	Emp_ID	Emp_Salary	Emp_Marks	Emp_Join_Date
0	1	90	63773	7	01-02-2015
1	2	15	86346	6	02-02-2015
2	3	22	71456	5	03-02-2015
3	4	43	60311	7	04-02-2015
4	5	25	70284	6	05-02-2015
5	6	55	60836	9	06-02-2015
6	7	14	52276	8	07-02-2015
7	8	29	79744	6	08-02-2015
8	9	46	65538	10	09-02-2015
9	10	9	77429	8	10-02-2015
10	11	47	97998	5	11-02-2015
11	12	55	87633	8	12-02-2015
12	13	79	85773	9	13-02-2015
13	14	61	53649	10	14-02-2015
14	15	41	74034	5	15-02-2015
15	16	7	64077	10	16-02-2015
16	17	45	57106	8	17-02-2015
17	18	58	58274	7	18-02-2015
18	19	72	97271	7	19-02-2015
19	20	65	52475	5	20-02-2015
20	21	60	92853	5	21-02-2015
21	22	43	93106	7	22-02-2015
22	23	72	99773	10	23-02-2015
23	24	68	60013	10	24-02-2015
24	25	8	88503	10	25-02-2015
25	26	40	84479	8	26-02-2015
26	27	44	51156	7	27-02-2015
27	28	9	58126	10	28-02-2015
28	29	54	79824	10	01-03-2015

29	30	20	97448	5	02-03-2015
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```
[23]: df.dropna(how="all")
```

```
[23]:
```

	Sr.No	Emp_ID	Emp_Salary	Emp_Marks	Emp_Join_Date
0	1	90	63773	7	01-02-2015
1	2	15	86346	6	02-02-2015
2	3	22	71456	5	03-02-2015
3	4	43	60311	7	04-02-2015
4	5	25	70284	6	05-02-2015
5	6	55	60836	9	06-02-2015
6	7	14	52276	8	07-02-2015
7	8	29	79744	6	08-02-2015
8	9	46	65538	10	09-02-2015
9	10	9	77429	8	10-02-2015
10	11	47	97998	5	11-02-2015
11	12	55	87633	8	12-02-2015
12	13	79	85773	9	13-02-2015
13	14	61	53649	10	14-02-2015
14	15	41	74034	5	15-02-2015
15	16	7	64077	10	16-02-2015
16	17	45	57106	8	17-02-2015
17	18	58	58274	7	18-02-2015
18	19	72	97271	7	19-02-2015
19	20	65	52475	5	20-02-2015
20	21	60	92853	5	21-02-2015
21	22	43	93106	7	22-02-2015
22	23	72	99773	10	23-02-2015
23	24	68	60013	10	24-02-2015
24	25	8	88503	10	25-02-2015
25	26	40	84479	8	26-02-2015
26	27	44	51156	7	27-02-2015
27	28	9	58126	10	28-02-2015
28	29	54	79824	10	01-03-2015
29	30	20	97448	5	02-03-2015

```
[25]: df.dropna(axis=1)
```

```
[25]:
```

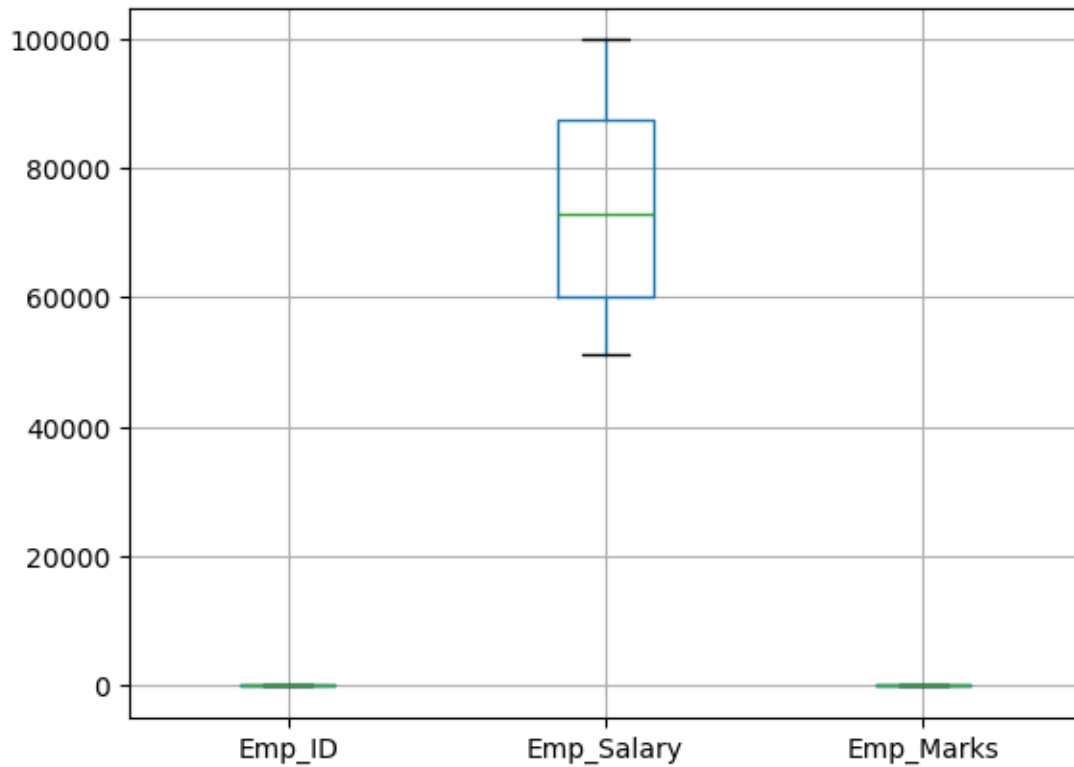
	Sr.No	Emp_ID	Emp_Salary	Emp_Marks	Emp_Join_Date
0	1	90	63773	7	01-02-2015
1	2	15	86346	6	02-02-2015
2	3	22	71456	5	03-02-2015
3	4	43	60311	7	04-02-2015
4	5	25	70284	6	05-02-2015
5	6	55	60836	9	06-02-2015
6	7	14	52276	8	07-02-2015
7	8	29	79744	6	08-02-2015

8	9	46	65538	10	09-02-2015
9	10	9	77429	8	10-02-2015
10	11	47	97998	5	11-02-2015
11	12	55	87633	8	12-02-2015
12	13	79	85773	9	13-02-2015
13	14	61	53649	10	14-02-2015
14	15	41	74034	5	15-02-2015
15	16	7	64077	10	16-02-2015
16	17	45	57106	8	17-02-2015
17	18	58	58274	7	18-02-2015
18	19	72	97271	7	19-02-2015
19	20	65	52475	5	20-02-2015
20	21	60	92853	5	21-02-2015
21	22	43	93106	7	22-02-2015
22	23	72	99773	10	23-02-2015
23	24	68	60013	10	24-02-2015
24	25	8	88503	10	25-02-2015
25	26	40	84479	8	26-02-2015
26	27	44	51156	7	27-02-2015
27	28	9	58126	10	28-02-2015
28	29	54	79824	10	01-03-2015
29	30	20	97448	5	02-03-2015

```
[27]: col = ['Emp_ID', 'Emp_Salary', 'Emp_Marks']
```

```
[29]: df.boxplot(col)
```

```
[29]: <Axes: >
```



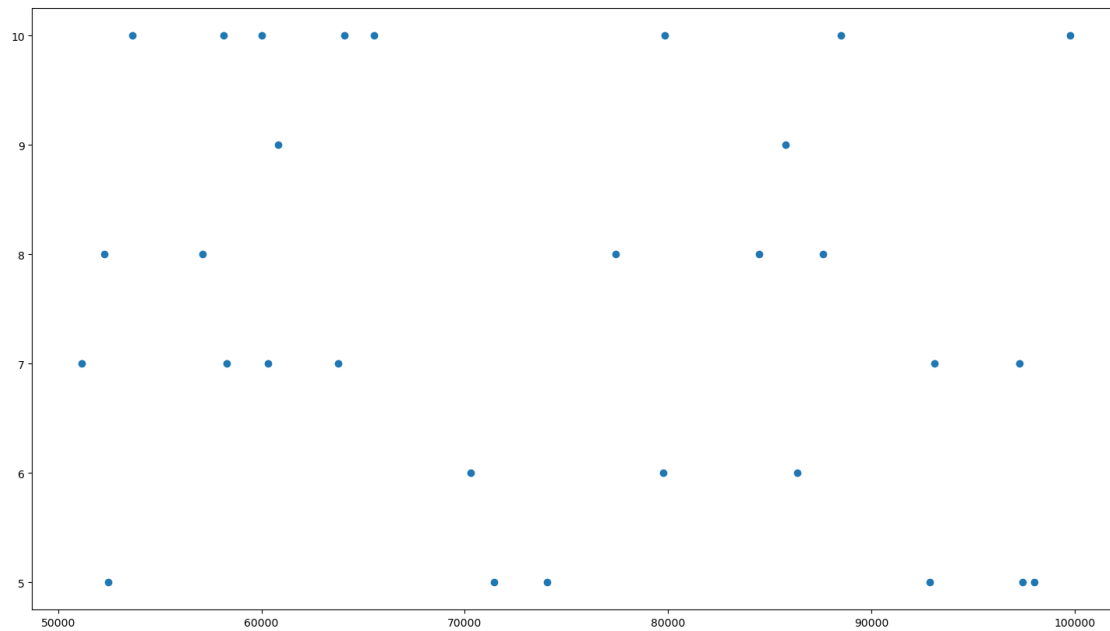
```
[30]: plt.show()
```

```
[33]: fig , ax =plt.subplots(figsize=(18,10))
```

```
[35]: ax.scatter(df['Emp_Salary'] , df['Emp_Marks'])
```

```
[35]: <matplotlib.collections.PathCollection at 0x29f7584c830>
```

```
[37]: plt.show()
```

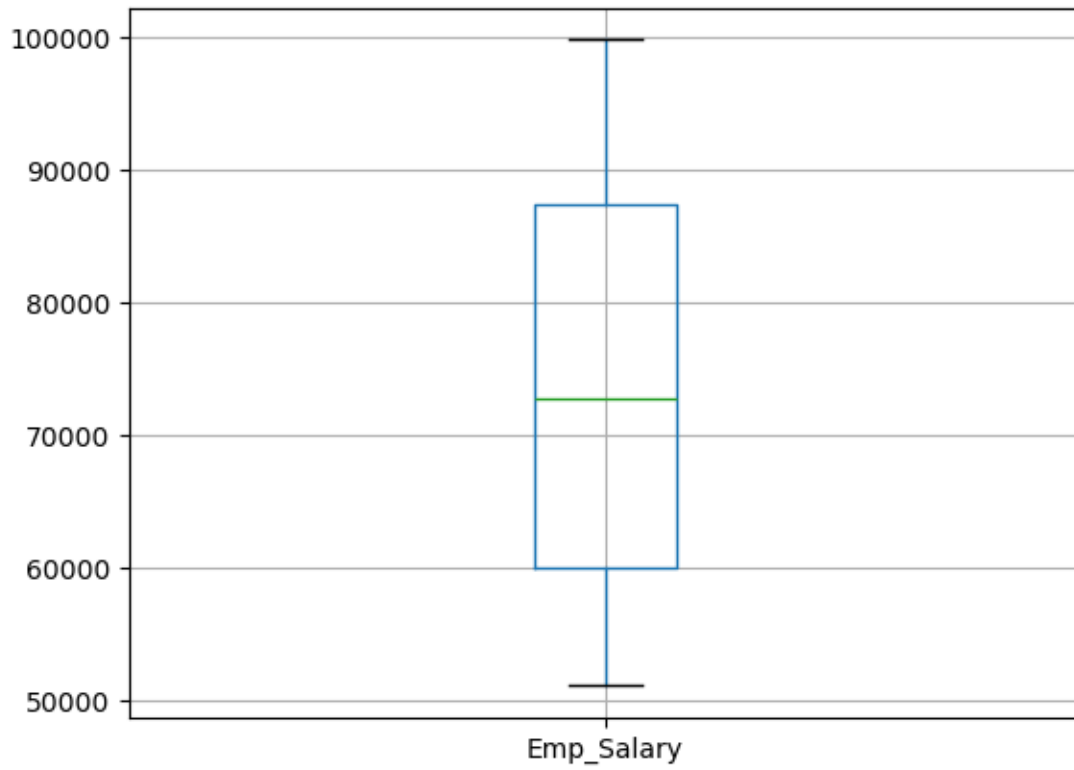


```
[39]: col = ['Emp_Salary']
```

```
[41]: df.boxplot(col)
```

```
[41]: <Axes: >
```

```
[43]: plt.show()
```

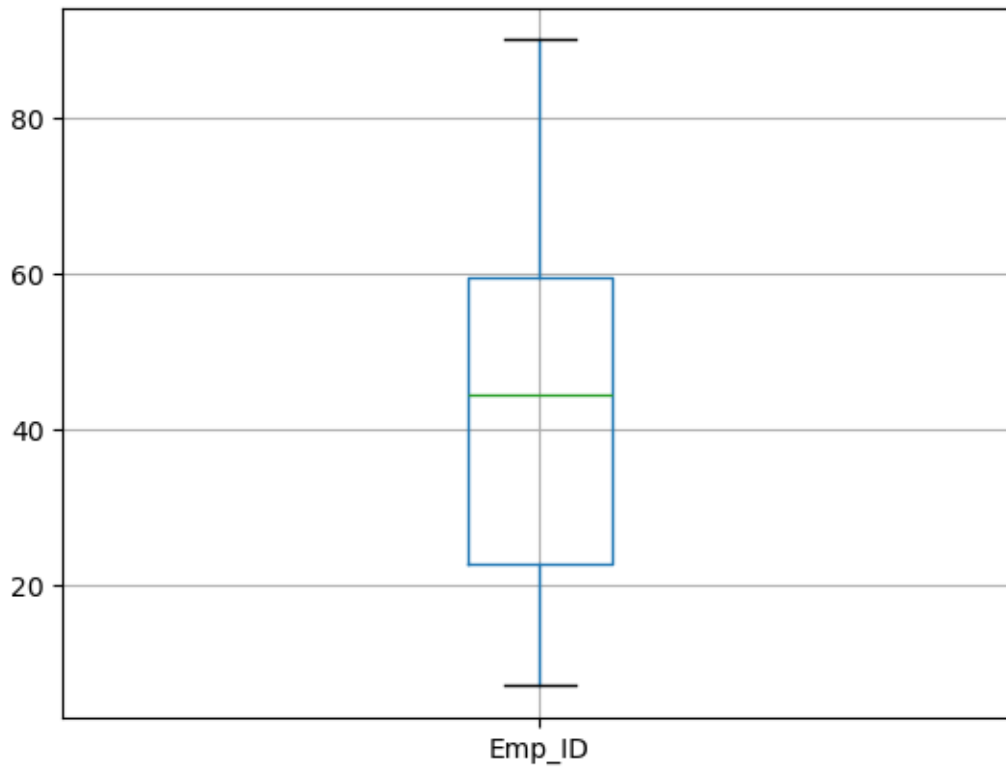



```
[45]: col = ['Emp_ID']
```

```
[47]: df.boxplot(col)
```

```
[47]: <Axes: >
```

```
[49]: plt.show()
```

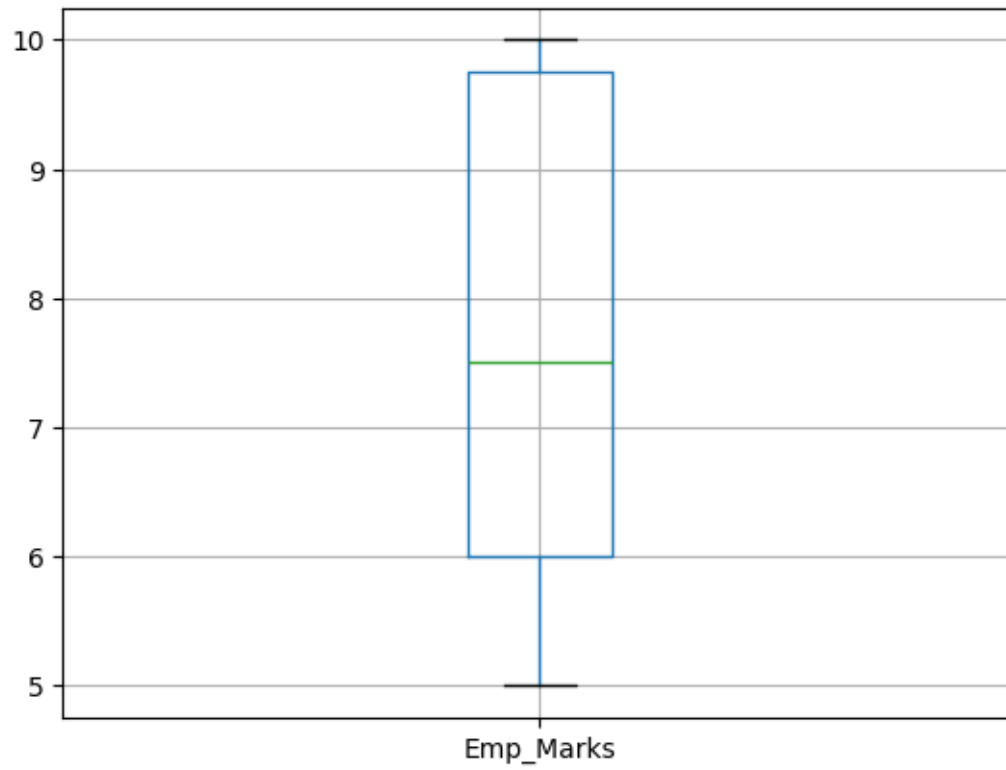


```
[51]: col = ['Emp_Marks']
```

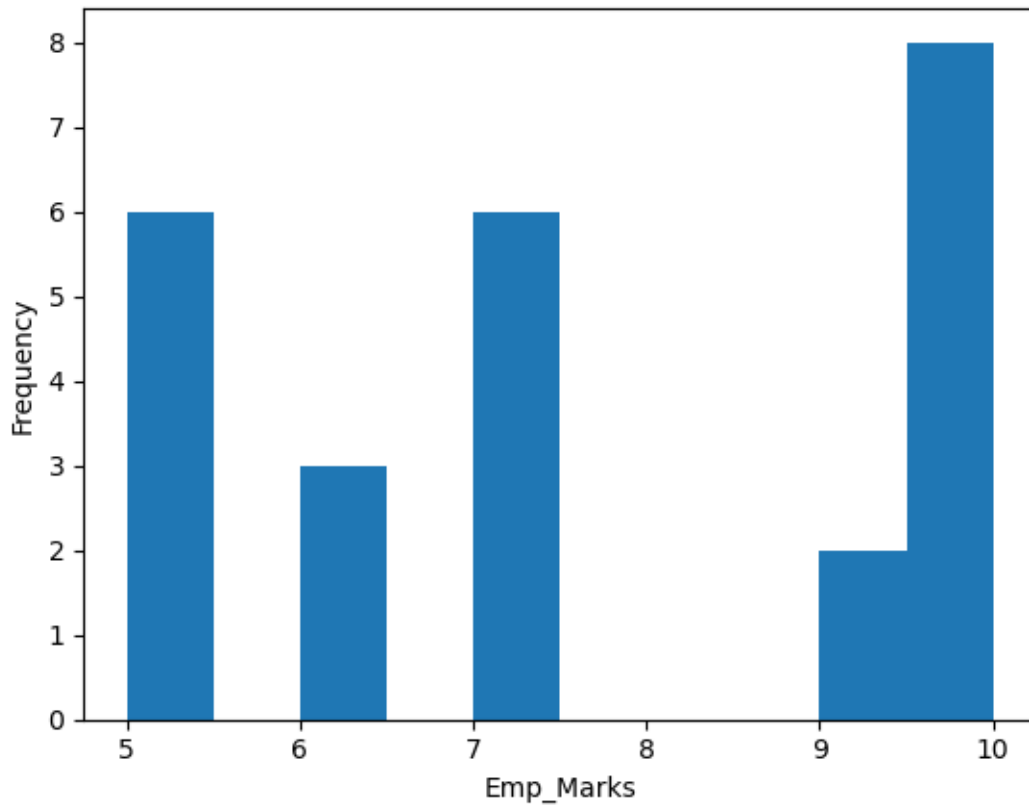
```
[53]: df.boxplot(col)
```

```
[53]: <Axes: >
```

```
[55]: plt.show()
```



```
[105]: df['Emp_Marks'].plot(kind='hist')
plt.xlabel('Emp_Marks')
plt.ylabel('Frequency')
plt.show()
```



```
[71]: indices=print(np.where(df['Emp_Marks']>7))
print(indices)

(array([ 5,  6,  8,  9, 11, 12, 13, 15, 16, 22, 23, 24, 25, 27, 28],
      dtype=int64),)
None
```

```
[73]: indices=print(np.where(df['Emp_Marks']<7))
print(indices)

(array([ 1,  2,  4,  7, 10, 14, 19, 20, 29], dtype=int64),)
None
```

```
[83]: from scipy import stats
z=np.abs(stats.zscore(df['Emp_Marks']))
print(z)

0    0.326679
1    0.871145
2    1.415610
3    0.326679
4    0.871145
```

```

5      0.762252
6      0.217786
7      0.871145
8      1.306717
9      0.217786
10     1.415610
11     0.217786
12     0.762252
13     1.306717
14     1.415610
15     1.306717
16     0.217786
17     0.326679
18     0.326679
19     1.415610
20     1.415610
21     0.326679
22     1.306717
23     1.306717
24     1.306717
25     0.217786
26     0.326679
27     1.306717
28     1.306717
29     1.415610
Name: Emp_Marks, dtype: float64

```

```

[91]: threshold = 0.22
      sample_outliers=np.where(z<threshold)
      sample_outliers

```

```

[91]: (array([ 6,  9, 11, 16, 25], dtype=int64),)

```

```

[99]: new_df = df.copy()

      # Dropping outliers from new_df
      for i in sample_outliers:
          new_df.drop(i, inplace=True, errors='ignore')

      # Display the updated new_df
      new_df

```

```

[99]:
   Sr.No  Emp_ID  Emp_Salary  Emp_Marks  Emp_Join_Date
0      1      90      63773          7    01-02-2015
1      2      15      86346          6    02-02-2015
2      3      22      71456          5    03-02-2015
3      4      43      60311          7    04-02-2015
4      5      25      70284          6    05-02-2015

```

5	6	55	60836	9	06-02-2015
7	8	29	79744	6	08-02-2015
8	9	46	65538	10	09-02-2015
10	11	47	97998	5	11-02-2015
12	13	79	85773	9	13-02-2015
13	14	61	53649	10	14-02-2015
14	15	41	74034	5	15-02-2015
15	16	7	64077	10	16-02-2015
17	18	58	58274	7	18-02-2015
18	19	72	97271	7	19-02-2015
19	20	65	52475	5	20-02-2015
20	21	60	92853	5	21-02-2015
21	22	43	93106	7	22-02-2015
22	23	72	99773	10	23-02-2015
23	24	68	60013	10	24-02-2015
24	25	8	88503	10	25-02-2015
26	27	44	51156	7	27-02-2015
27	28	9	58126	10	28-02-2015
28	29	54	79824	10	01-03-2015
29	30	20	97448	5	02-03-2015

```
[57]: from sklearn.preprocessing import LabelEncoder
      le=LabelEncoder()
```

```
[59]: df['Emp_Join_Date']=le.fit_transform(df['Emp_Join_Date'])
      newdf=df
```

```
[61]: df
```

```
[61]:
```

	Sr.No	Emp_ID	Emp_Salary	Emp_Marks	Emp_Join_Date
0	1	90	63773	7	0
1	2	15	86346	6	2
2	3	22	71456	5	4
3	4	43	60311	7	5
4	5	25	70284	6	6
5	6	55	60836	9	7
6	7	14	52276	8	8
7	8	29	79744	6	9
8	9	46	65538	10	10
9	10	9	77429	8	11
10	11	47	97998	5	12
11	12	55	87633	8	13
12	13	79	85773	9	14
13	14	61	53649	10	15
14	15	41	74034	5	16
15	16	7	64077	10	17
16	17	45	57106	8	18

17	18	58	58274	7	19
18	19	72	97271	7	20
19	20	65	52475	5	21
20	21	60	92853	5	22
21	22	43	93106	7	23
22	23	72	99773	10	24
23	24	68	60013	10	25
24	25	8	88503	10	26
25	26	40	84479	8	27
26	27	44	51156	7	28
27	28	9	58126	10	29
28	29	54	79824	10	1
29	30	20	97448	5	3

```
[65]: df['Emp_Join_Date'] = le.inverse_transform(df['Emp_Join_Date'])
newdf=df
df
```

```
[65]:
```

	Sr.No	Emp_ID	Emp_Salary	Emp_Marks	Emp_Join_Date
0	1	90	63773	7	01-02-2015
1	2	15	86346	6	02-02-2015
2	3	22	71456	5	03-02-2015
3	4	43	60311	7	04-02-2015
4	5	25	70284	6	05-02-2015
5	6	55	60836	9	06-02-2015
6	7	14	52276	8	07-02-2015
7	8	29	79744	6	08-02-2015
8	9	46	65538	10	09-02-2015
9	10	9	77429	8	10-02-2015
10	11	47	97998	5	11-02-2015
11	12	55	87633	8	12-02-2015
12	13	79	85773	9	13-02-2015
13	14	61	53649	10	14-02-2015
14	15	41	74034	5	15-02-2015
15	16	7	64077	10	16-02-2015
16	17	45	57106	8	17-02-2015
17	18	58	58274	7	18-02-2015
18	19	72	97271	7	19-02-2015
19	20	65	52475	5	20-02-2015
20	21	60	92853	5	21-02-2015
21	22	43	93106	7	22-02-2015
22	23	72	99773	10	23-02-2015
23	24	68	60013	10	24-02-2015
24	25	8	88503	10	25-02-2015
25	26	40	84479	8	26-02-2015
26	27	44	51156	7	27-02-2015
27	28	9	58126	10	28-02-2015

28	29	54	79824	10	01-03-2015
29	30	20	97448	5	02-03-2015

[]: